

Problem

Given that

$$\frac{dv}{dt} + 2v + 5 \int_0^t v(\lambda) d\lambda = 4u(t)$$

with $v(0) = -1$, determine $v(t)$ for $t > 0$.

Step-by-step solution

Step 1 of 2

Given differential equation

$$\frac{dv}{dt} + 2v + 5 \int_0^t v(\lambda) d\lambda = 4u(t)$$

By taking Laplace Transform on both sides for the above equation we can get

$$sV(s) - v(0) + 2V(s) + \frac{5}{s}V(s) = \frac{4}{s}$$

$$V(s) \left(s + 2 + \frac{5}{s} \right) = \frac{4}{s} + v(0)$$

$$V(s) \left(s + 2 + \frac{5}{s} \right) = \frac{4}{s} - 1$$

$$V(s) (s^2 + 2s + 5) = 4 - s$$

Step 2 of 2

$$V(s) = \frac{4-s}{s^2+2s+5}$$

$$V(s) = \frac{4-s}{(s^2+2s+1)+4}$$

$$V(s) = \frac{4-s}{(s+1)^2+2^2}$$

$$V(s) = \frac{4}{(s+1)^2+2^2} - \frac{s}{(s+1)^2+2^2}$$

$$V(s) = \frac{4}{(s+1)^2+2^2} - \frac{s+1-1}{(s+1)^2+2^2}$$

$$V(s) = \frac{4}{(s+1)^2+2^2} - \frac{s+1}{(s+1)^2+2^2} + \frac{1}{(s+1)^2+2^2}$$

$$V(s) = \frac{5}{(s+1)^2+2^2} - \frac{s+1}{(s+1)^2+2^2}$$

$$V(s) = \left(\frac{5}{2}\right) \left(\frac{2}{(s+1)^2+2^2}\right) - \frac{s+1}{(s+1)^2+2^2} \text{ By taking inverse Laplace Transform on both sides}$$

we can get

$$v(t) = \left(\frac{5}{2}\right) e^{-t} \sin(2t) u(t) - e^{-t} \cos(2t) u(t)$$

$$\boxed{v(t) = \left[\frac{5}{2} \sin(2t) - \cos(2t)\right] e^{-t} u(t)}$$